[Aim: 100/100 in Maths]

आरंभ

REAL NUMBERS

Lecture #6

#ATK²:-

→ LCM & HCF by $\boxed{\text{prime factorisation}}$

→ Prime & composite No.

→ p & q → prime → HCF = 1, LCM = $p \times q$

→
$$\begin{tikzcd}
\text{HCF} \arrow[r, bend left, "\text{factor}"] & \text{LCM} \arrow[l, bend left, "\text{multiple}"]
\end{tikzcd}$$

→ $\boxed{HCF \times LCM = a \times b}$ for $\boxed{\text{two}}$ numbers

3 Nos
4 Nos
5 Nos } not true

$H \times L \neq a \times b \times c$

$8 = \overbrace{2 \times 2 \times 2}^{\text{prime}}$

$24 = \underbrace{2 \times 2 \times 2 \times 3}_{\text{prime}}$

$HCF = 2 \times 2 \times 2$

$\cancel{8 = 4 \times 2}$

$\cancel{24 = 8 \times 3}$

$\cancel{\boxed{HCF = 1}}$

prime no. मे तोड़ना है then common लेना है !!

#LP: P=2(4)(6).....(20) and Q=1(3)(5).....(19). What is the HCF of P and Q?

$P = 2 \times 4 \times 6 \times 8 \times 10 \times 12 \times 14 \times 16 \times 18 \times 20$

$P = 2 \times 2 \times 2 \times 2 \times 3 \times 2 \times 2 \times 2 \times 5 \times 2 \times 2 \times 2 \times 3 \times 2 \times 7 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 2 \times 2 \times 5$

$\boxed{P = 2^{18} \times 3^4 \times 5^2 \times 7^1}$ ← prime factorisation/product of prime

$Q = 1 \times 3 \times 5 \times 7 \times 9 \times 11 \times 13 \times 15 \times 17 \times 19$

$Q = \underbrace{\qquad\qquad}$

$8 = \underbrace{2 \times 2 \times 2}_{\text{prime no.}}$

composite

$26 = \underbrace{2 \times 13}_{\text{product of prime}}$

composite

$20 = \underbrace{2 \times 2 \times 5}_{\text{prime.}}$

composite

#FUNDAMENTAL THEOREM OF ARITHMETIC:

According to the fundamental theorem of arithmetic, every composite number can be written (factorise) as the product of primes and this factorisation is unique, apart from the order in which the prime factors occur.

$$\underset{\text{Comp.}}{12} = \overbrace{2 \times 2 \times 3}^{\text{product of prime}}$$

$2 \times 3 \times 2$

$3 \times 2 \times 2$

#LP: Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers.

(let) $x \Rightarrow 7 \times 11 \times 13 + 13$

$x \Rightarrow 13(7 \times 11 + 1)$

$x \Rightarrow 13(77 + 1)$

$x \Rightarrow 13(78)$

1 ...

13 78 13×78

more than 2 factors.

$\therefore x$ is a composite no.

Next Toppers

#LP: Explain why ~~7 x 11 x 13 + 13~~ and 7 x 6 x 5 x 4 x 3 x 2 x 1 + 5 are composite numbers.

(let) $y = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

$= 5(7 \times 6 \times 4 \times 3 \times 2 \times 1 + 1)$

$\boxed{y = 5(1009)}$

5 1009 5(1009) 1

more than 2 factors

$\therefore$ it's a composite no.

#LP: Let p, q and r be three distinct prime numbers. Check whether $\boxed{pqr + q}$ is a composite number or not.
Further, give an example for three distinct primes p, q, r such that:
(i) pqr + 1 is a composite number
(ii) pqr + 1 is a prime number. [CBSE 2025]

$p, q, r \longrightarrow$ prime no.s

To check :- $(pqr + q)$ is composite or not?

$pqr + q$

$q(pr + 1)$

1

$q, (pr+1)$

q

$(pr+1)$

more than 2 factors

$\therefore$ $pqr + q$ is composite no!

Next Toppers

#LP: Let p, q and r be three distinct prime numbers. Check whether pqr + q is a composite number or not.
Further, give an example for three distinct primes p, q, r such that:
(i) pqr + 1 is a composite number
(ii) pqr + 1 is a prime number. **[CBSE 2025]**

$p, q, r \longrightarrow$ prime no.s

(i) $pqr+1 \longrightarrow$ composite no.

3 5 7

$p = 3$
$q = 5$
$r = 7$

$105+1$

$106 \longrightarrow$ composite

(ii) $(pqr+1) \longrightarrow$ prime no

2 3 5

$p = 2$
$q = 3$
$r = 5$

$pqr+1$

$2\times3\times5+1$

$31 \longrightarrow$ prime no.

Next Toppers

$$pqr + (q \times 1)$$

$$q(pr \times 1 + 1)$$

$$\boxed{q(pr+1)}$$

#LP: Show that 6^n cannot end with digit 0 any natural number n.

$(6)^n = (2 \times 3)^n$

$n=1 \rightarrow 6^1 = (2 \times 3)^1 = 2 \times 3$ 2 ✓ 5 ✗

$n=2 \rightarrow 6^2 = (2 \times 3)^2 = \underline{(2 \times 3)(2 \times 3)}$ 2 ✓ 5 ✗

$n=3 \rightarrow 6^3 = (2 \times 3)^3 = \underline{2 \times 3 \times 2 \times 3 \times 2 \times 3}$ 2 ✓ 5 ✗

$n=10 \rightarrow 6^{10} = (2 \times 3)^{10} = \underbrace{2 \times 3 \times 2 \times 3 \times 2 \times 3 \times 2 \times 3 \ldots}_{}$ 10 times 2 ✓ 5 ✗

$n=n \rightarrow 6^n = (2 \times 3)^n = \underbrace{2 \times 3 \times 2 \times 3 \times 2 \times 3 \ldots}_{n \text{ times}}$ 2 ✓ 5 ✗

K 3 B

for any number to end with 0.

_ _ _ _ 0

it's prime factorisation must contain 2 & 5 both.

Since prime factorisation of 6^n doesn't contain 2 and 5 both.

Hence, 6^n cannot end with 0.

#LP: Show that 9^n cannot end with digit 0 any natural number n.

$$9^n = (3\times3)^n$$

$n=1 \rightarrow 9^1 = (3\times3)^1$ $2\times$ $5\times$

$n=2 \rightarrow 9^2 = (3\times3)^2 = 3\times3\times3\times3$ $2\times$ $5\times$

$n=10 \rightarrow 9^{10} = (3\times3)^{10} = \underbrace{3\times3\times3\times3\times3\times3 - - - -}_{}$ 10 times $2\times$ $5\times$

$n=n \rightarrow 9^n = (3\times3)^n = \underbrace{3\times3\times3\times3\times3\times3 - - - - - -}_{}$ n times $2\times$ $5\times$

Next Toppers

Since prime factorisation of 9^n doesn't contain 2 & 5 both.

$\therefore$ 9^n cannot end with 0.

नाम पर मत जाना!!

Co-prime Numbers:

6: 1, 2, 3, 6

13: 1, 13

common factor = 1

or $\boxed{HCF = 1}$

Co-prime numbers

all prime no.s are coprime.

but all coprimes are not prime.

8: 1, 2, 4, 8 — not prim

15: 1, 3, 5, 15 — not prime

HCF = 1

∴ 8 & 15 are coprime nos.

5: 1, 5

7: 1, 7

HCF = 1

#LP: The sum of two numbers is 1215 and their HCF is 81, then the possible pairs of such numbers are. [CBSE SQP 2025-26]

(A) 2
(B) 3
(C) 4
(D) 5

Let the no.s are $81x$ & $81y$, x & y are coprime.

ATQ

$\text{sum} = 1215$

$81x + 81y = 1215$

$81(x+y) = 1215$

$x + y = \cancel{\frac{1215}{81}}\ \cancel{\frac{405}{27}}\ \cancel{\frac{135}{9}}\ \cancel{\frac{45}{3}}\ 15$

$\boxed{x+y=15}$

$x+y=15$

✓ $\boxed{(1,14)}$
✓ $\boxed{(2,13)}$
✗ $(3,12)$
✓ $\boxed{(4,11)}$
✗ $(5,10)$
✗ $(6,9)$
✓ $\boxed{(7,8)}$

$\therefore$ 4 pairs

Ans

#LP: The number of possible pairs such that the product of two numbers and HCF are 4500 and 15 respectively:

A. 1

~~B. 2~~ ✓

C. 3

D. 4

$\boxed{HCF = 15}$

Let the two nos are $15x$ & $15y$ ∴ x & y are co-prime

ATQ Product = 4500

$15x \times 15y = 4500$

$x \times y = \dfrac{\cancel{4500}}{\cancel{15} \times \cancel{15}}$ 20

$\boxed{x \cdot y = 20}$

$x \cdot y = 20$

✓ $\boxed{(1, 20)}$

✓ $\boxed{(4, 5)}$

✗ $(10, 2)$

∴ 2 pairs

No HW

THANK YOU COODIESS !!

Class 10^{th} Aarambh 2026-27
DPP #3

#LP 1 : Assertion(A): **The H.C.F. of two numbers is 16 and their product is 3072. Then their L.C.M. = 162.**
Reason(R): **If a and b are two positive integers, then H.C.F. × L.C.M. = a × b.**
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of
Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

#LP 2: Assertion(A): **For any two positive integers p and q, HCF (p, q) × LCM (p, q) = p × q**
Reason(R): **If the HCF of two numbers is 5 and their product is 150, then their LCM is 40.**
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of
Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

#LP 3: The least number which is a perfect square and is divisible by each of 16, 20 and 24 is
(A) 240
(B) 1600
(C) 2400
(D) 3600

#LP 4: If the product of two numbers is 1050 and their HCF is 25, find their LCM.

(A) 42
(B) 25
(C) 105
(D) none of these

#LP 5: The largest number which divides 70 and 125, leaving remainders 5 and 8, respectively, is

(A) 13
(B) 65
(C) 875
(D) 1750

#LP 6: If HCF (98, 28) = m and LCM (98, 28) = n, then the value of n - 7m is:

(A) 0
(B) 28
(C) 98
(D) 198

#LP 7: Write the smallest number which is divisible by both 306 and 657.

#LP8:Fill in the blanks:

(a) If every positive even integer is of the form 2q, then every positive odd integer is of the form ______, where q is some integer.
(b) The exponent of 2 in the prime factorisation of 144 is ______.
(c) If p is a prime number and it divides a^2, then it also divides ______, where a is a positive integer.
(d) Every point on the number line corresponds to a ______ number.
(e) The product of three numbers is ______ to the product of their HCF and LCM.

Part - 2 (Advanced Subjective Type Questions)

1. Prime factorisation of three numbers A, B and C is given below :

$$A = \left(2^r \times 3^p \times 5^q\right)$$

$$B = \left(2^p \times 3^r \times 5^p\right)$$

$$C = \left(2^p \times 3^q \times 5^p\right)$$

such that , $p < q < r$ and p, q, and r are natural numbers.

- The largest number that divides A, B and C without leaving a remainder is 30.
- The smallest number that leaves a remainder of 2 when divided by each of A, B and C is 5402

Find A, B and C. Show your work.

4. P is the LCM of 2, 4, 6, 8, 10; Q is the LCM of 1, 3, 5, 7, 9 and L is the LCM of P and Q. Then, find the relation between L and P.

5. On the two real numbers $a = 2 + \sqrt{5}$ and $b = 3 - \sqrt{7}$, perform the following operations :

[CBSE CFPQ]

(i) Calculate the sum (a + b)

(ii) Calculate the product (ab)

(iii) Find the additive inverse of a

(iv) Rationalise 1/b

(v) Verify whether the numbers a and b are rational or irrational. Provide a valid reason for your answer

Part - 3 (Advanced Miscellaneous Questions)

6. Match the Column I with Column II and choose the correct option :

Column-I	Column-II
(A) The HCF of the smallest number of two digits and the largest multiple of 5 which is less than 40 is	(i) 5
(B) HCF of $x^3y^4z^2$ and $x^2y^3z^5$, where x, y and z are distinct prime numbers is	(ii) 4
(C) The total number of factors of a prime number is	(iii) 2
(D) The sum of exponents of prime factors in the prime factorisation of 196 is	(iv) $x^2y^3z^2$

(a) A-iv, B-ii, C-i, D-iii

(b) A-i, B-iv, C-iii, D-ii

(c) A-ii, B-iv, C-i, D-iii

(d) A-ii, B-iv, C-iii, D-i